WADII BOULILA

Professor of Computer Science | Director, RIOTU Laboratory | Prince Sultan University, Saudi Arabia

Trustworthy AI · Computer Vision · Remote Sensing · IoT & Edge Intelligence · LLMs & NLP · Uncertainty Modeling · Data Fusion

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Research Vision & Trajectory

My research programme is centred on the design of **trustworthy intelligent systems for complex, data-rich, and uncertainty-prone environments**. It builds on foundational work in **uncertainty modelling, information fusion, and knowledge discovery from geospatial data** (Ph.D., Rennes 1; HDR, Manouba), and has evolved toward **scalable deep learning, computer vision, privacy-preserving AI, federated learning, and IoT security** for real-world applications. Across these themes, the unifying goal of my research is to develop intelligent systems that can operate reliably under uncertainty, at scale, and under resource and privacy constraints. More recently, my work has increasingly moved toward the convergence of **edge intelligence, multimodal AI, and trustworthy autonomous systems**, with growing interest in emerging paradigms such as **agentic AI** for distributed and adaptive decision-making. As Director of the RIOTU Laboratory, I am advancing a research agenda that integrates **remote sensing, IoT, machine learning, computer vision, and language-enabled AI** to develop robust and deployable solutions for precision agriculture, environmental monitoring, intelligent healthcare, and autonomous unmanned systems. The long-term goal of this trajectory is to bridge methodological advances in AI with high-impact operational challenges, while building the scientific infrastructure, international partnerships, and translational pathways needed to support internationally visible and societally relevant research.

Citation Metrics & Research Impact

Publications	Citations	h-index	i10-index	Q1 Journals	Books
300+	9400+	53	147	110 / 160	5
Authorship	**Sole Author**	**1st Author**	**2nd Author**	**Last**	**Other**
	6	30	60	60	154

World's Top 2% Scientist (Stanford/Elsevier, 2022–2025, four consecutive years). As of March 2026.

Signature Research Contributions

1. **Uncertainty Modelling and Geospatial Intelligence.** Established a sustained research line on **uncertainty-aware modelling, information fusion, and knowledge discovery from satellite and geospatial data** (Ph.D., Rennes 1; HDR, Manouba). This work provided the methodological foundation for my later contributions to trustworthy AI, with a focus on how intelligent systems interpret imperfect, heterogeneous, and evolving spatiotemporal information.

2. **Scalable Deep Learning and Computer Vision.** Advanced **distributed and high-capacity deep learning architectures** for large-scale visual analytics in remote sensing and image-based applications. These contributions addressed practical bottlenecks in memory, scale, and modelling complexity, strengthening the link between advanced AI methodology, **computer vision**, and operational geospatial analysis.

3. **Privacy-Preserving and Trustworthy AI for Sensitive Data.** Contributed to the development of **privacy-aware, federated, and secure learning pipelines** for sensitive datasets, particularly in healthcare diagnostics and remote sensing, showing that decentralization and privacy constraints can be incorporated without compromising practical utility. This line connects naturally with my broader work on **IoT security, distributed intelligence, and dependable AI systems** for cyber–physical environments.

4. **Translational AI for Precision Agriculture and Intelligent Monitoring.** Developed AI-driven solutions for agricultural and field environments, most notably through deep-learning-based early detection of the **Red Palm Weevil** from acoustic signals and UAV-supported monitoring. This work led to a **Saudi patent (SAIP SA 18938 B1)** and reflects my broader commitment to translating AI research into robust systems for precision agriculture, environmental monitoring, and autonomous sensing platforms.

5. **Emerging Intelligent Systems: LLMs, Generative AI, and Autonomous Platforms.** More recently, my research has expanded toward the convergence of **large language models, generative AI, edge intelligence, and autonomous unmanned systems**, focusing on how foundation-model capabilities can support perception, reasoning, and adaptive decision-making under operational and resource constraints. Complementing this direction, I have authored and co-authored **widely cited survey and synthesis papers** spanning IoT, graph deep learning, LLMs, and generative AI, helping to consolidate and connect emerging interdisciplinary research areas.

Education

- **Qualification to lead research in Computer Science** *2019*
  National College of Computer Science, University of Manouba Tunisia
  ◦ *From Remote Sensing Data to Knowledge: Modelling Uncertainties*
- **Ph.D. in Computer Science** *2009 – 2012*
  University of Rennes 1 France
  ◦ *Spatiotemporal Uncertain Knowledge Discovery for Land Cover Change Prediction*
- **M.Sc. in Computer Science** *2005 – 2007*
  National College of Computer Science, University of Manouba Tunisia
  ◦ *A Multi-Approach System for Satellite Image Interpretation via Fusion of Uncertain Information*
- **B.Sc. in Computer Science (First Class Honours)** *2000 – 2005*
  Aviation College of Borj El Amri, Tunis University Tunisia
  ◦ Highest First-Class Honours three consecutive years (2003, 2004, 2005). Best Graduation Project Award (2005).
- **Baccalaureate in Mathematics (Top 1% in Tunisia)** *2000*
  School of El Haouaria Tunisia

Academic Appointments

- **Professor of Computer Science** *2025 – Present*
  Prince Sultan University, CCIS Riyadh, Saudi Arabia
- **Director, RIOTU Laboratory (20+ members; largest lab at PSU)** *2024 – Present*
  Prince Sultan University Riyadh, Saudi Arabia
- **Associate Professor of Computer Science** *2022 – 2025*
  Prince Sultan University, CCIS Riyadh, Saudi Arabia
- **Senior Researcher, RIOTU Laboratory** *2021 – 2022*
  Prince Sultan University, CCIS Riyadh, Saudi Arabia
- **Associate / Assistant Professor of Computer Science** *2015 – 2021*
  Taibah University, CCSE Madinah, Saudi Arabia
- **Assistant Professor of Computer Science** *2012 – 2015*
  ISAMM, Manouba University Tunisia
- **Dept. Chair; Director of Maintenance Center; Director of Application Development Center** *2007 – 2012*
  Aviation College of Borj El Amri Tunisia
- **Network and Systems Engineer** *2005 – 2007*
  Aviation College of Borj El Amri Tunisia

Academic Leadership & Institution-Building

- **Director, RIOTU Laboratory** (2024–present): Founded and scaled PSU's first and largest research lab to 20+ full-time members (senior researchers, postdocs, engineers, graduate students). Secured competitive multi-agency funding, built partnerships with academia, industry, and government aligned with Saudi Vision 2030. Awarded *Highest Impact Factor Research Lab* (PSU, 2023).
- **M.Sc. & B.Sc. in AI Curriculum Architect** (2023–2026): Core member of the Academic Committee for Establishing a Master and Bachelor Programs in Artificial Intelligence at PSU—contributing to programme design, course sequencing, learning outcomes, and ABET-aligned assessment.
- **Conference & Community Builder:** General Chair of 3 international conferences (SmartTech 2022, 2024; IPCAINT 2024). Organized 20+ special sessions at IJCNN, KES, ICCCI, IWCMC, IRICT. Built sustainable partnerships bringing international speakers and co-investigators.
- **Departmental Leadership:** Chairman of Graduation Projects, IT, and Administrative Committees at Taibah University (2015–2021). Coordinator of the BSc programme at ISAMM (2012–2015). Department Chair at EABA (2010–2012).

Honors & Awards

- **World's Top 2% Scientist** (Stanford University / Elsevier) *2022, 2023, 2024, 2025*
- **Premium Residency—Exceptional Talents Category** Kingdom of Saudi Arabia *2024*
- **Global Talent Endorsement** University of Manchester, UK *2022*
- **Best Young Researcher in CS, Tunisia** Beit El Hikma—Tunisian Academy of Sciences *2021*
- **Best Researcher (All Fields) + Most Cited Researcher** University of Manouba *2021, 2022*
- **Research Excellence Award** Prince Sultan University *2023*
- **Climate Change Award** Prince Sultan University *2023*
- **Top 3 Faculty for Most Impactful Research Output** CCIS, Prince Sultan University *2024, 2025*
- **Outstanding Teacher Award** CCIS, Prince Sultan University *2025*
- **2nd Place, CVPR 2025 Night Photography Challenge** (top-ranked AI conference) *2025*
- **2nd Place, ROSHN Hackathon—AI & Automation** (1,800 participants) *2025*
- **Top-Cited Paper Awards** Comp. Sci. Rev. , Comp. Biol. Med. , IJIST, Softw. Pract. Exper. *2021–2025*
- **Best Paper Awards** SmartTech 2024, ICACIn 2021, ICOICS 2021 *2021–2024*
- **Best Assistant Professor in Tunisia** *2012*

Research Funding

Funding overview: 1.585M SAR (~422K USD) across **11 competitive grants** from 7 distinct funding agencies. **4 grants as PI** (290K SAR), **7 as Co-PI** (1.295M SAR). Funding secured from the Saudi Ministry of Education (competitive national call; acceptance rate ~15–20%), the UK Global Challenges Research Fund, King Abdulaziz University, Prince Sultan Defense Studies and Research Center (PSDSARC), Prince Sultan University, and the Saudi Ministry of Culture (competitive national call; acceptance rate 4%). As RIOTU Director, currently managing a multi-grant, multi-investigator research portfolio positioning the lab for larger-scale international consortia (e.g., Horizon Europe, KACST National Priorities).

Role	Project	Funder	SAR	Period
Co-PI	AI Copyright Protection for Publishing Houses	Ministry of Culture	35K	2026–27
PI	QuadroRL: Search & Rescue Quadrupeds	PSU Seed	30K	2025–26
PI	SkyHarvest: Palm Analytics via Domain Adaptation	PSU Seed	30K	2023–24
Co-PI	Terrain-Aided UAV Navigation in GPS-Denied Env.	PSDSARC	300K	2023–24
PI	ServePalms: Smart Palm Tracking & Disease Detection	PSU Seed	30K	2022–23
Co-PI	Deep Big Data for Environmental Services	King Abdulaziz University	20K	2022–23
Co-PI	Smart Healthcare for Brain Disorders	Ministry of Education	350K	2021–23
Co-PI	SmartWater: Real-Time IoT Water Monitoring	Ministry of Education	350K	2020–22
PI	COVID-19 Respiratory Disorder Detection	Taibah University	200K	2020–21
Co-PI	Health-Related Rumor Detection for COVID-19	Taibah University	200K	2020–21
Co-PI	Secure Pop-Up Networks	GCRF/SFC (UK)	40K	2019

Patents & Intellectual Property

[P.1] Patent (SAIP, SA 18938 B1): Detection of Red Palm Weevil Infestation in Palm Farms, 2024.

[P.2] Copyright (SAIP, 25-12-47001624): Attention-Enhanced Dual-Branch Network for Dynamic Light Enhancement, 2025.

Graduate Supervision

Ph.D.: 10 (6 completed, 4 in progress) across Tunisia, France, UK, Saudi Arabia **M.Sc.:** 25+ **B.Sc.:** 50+
External Examiner: 9 theses internationally (including Università di Bologna, Italy; King Saud University, KSA; King Abdulaziz University, KSA; and VIT-AP University, India).

Completed doctoral students:

1. A. Ferchichi — Uncertainty in Land Cover Models (U. Manouba, 2017)
2. M.W. Farouq — Gene Expression Modeling (U. Stirling, UK, 2018)
3. I. Chebbi — Spatiotemporal Big Remote Sensing (U. Paris 8 / U. Manouba, 2020)
4. Z. Ayadi — Segmentation via Uncertain Constraints (U. Strasbourg / U. Manouba, 2022)
5. S.B. Atitallah — Intelligent Microservices for IoT (U. Manouba, 2023)
6. Y. Hajjaji — IoT & DL for Remote Sensing (Edinburgh Napier / U. Manouba, 2023)

Current: F. Yakoubi (U. Carthage, 2021–), M. Khazri (U. Manouba, 2022–), O. Belhwichet (U. Carthage, 2022–), N. Ben Ameur (UTBM France / U. Tunis El Manar, 2026–)

International Academic Affiliations & Professional Memberships

- **IEEE Senior Member**
- **Senior Fellow, Higher Education Academy (SFHEA)**, UK
- **Member, AAAI**— Association for the Advancement of Artificial Intelligence
- **Member, ACM**
- **Adjunct Professor**, VIT University, India (Ranked #1 Private Engineering)
- **Founding Head**, Data Science Group, Manouba University, Tunisia
- **Senior Researcher**, RIADI Laboratory, University of Manouba, Tunisia (2006–present)
- **Senior Research Fellow**, ITI Dept., University of Rennes 1, France (2015–2020)
- **Senior Research Fellow**, CogBID Laboratory, University of Stirling, UK (2015–2019)

Invited & Keynote Addresses (Selected)

"Bridging the AI Implementation Gap: From Real Needs to Scaled Impact" *2026*
3rd IBI, Tunisia

"Modern Trends in Artificial Intelligence" *2026*
Digital Government Authority Workshop, Saudi Arabia

"The Data-Efficient Path to Action: Self-Supervised Pretraining for Large Action Models" *2025*
4th OCTA, Tunisia

"Teaching Machines to Act: Large Action Models and the Era of Agentic AI " *2025*
4th SmartTech, Tunisia

"Deep Learning at Scale for Remote Sensing and IoT" *2025*
Distinguished Speaker, Jinan Univ. & Jiangsu Univ. of Sci. & Tech., China

"Minimal Data, Maximum Impact: Optimizing Deep Learning for Real-World Applications" *2025*
University of Sharjah Alumni Conference, UAE

"AI-Powered Early Detection System for Red Palm Weevil" *2025*
University of Technology Bahrain; Bahrain

"Agentic AI: A New Era of Autonomous and Sustainable Intelligence" *2025*
Prince Sultan University; Saudi Arabia

"Rethinking Machine Intelligence: The Rise of Few-Shot Learning" *2024*
2nd IPCAINT, Morocco

"Reimagining the Future of Computer Vision: From Object Detection to Generative Models, and Beyond" *2023*
7th IRICT, Malaysia

"Big Remote Sensing Data Analytics and Uncertainty Modeling" *2022*
King Abdullah University of Science and Technology (KAUST), Saudi Arabia

Editorial & Conference Service

Associate Editor: Frontiers in Communications & Networks — IoT and Sensor Networks (2022–present); Int. J. Business Data Comm. & Networking (2021–present).
Guest Editor: IEEE J. Biomedical & Health Informatics, IF 7.7 (2024–2025); Int. J. Remote Sensing, IF 3.0 (2024–2025); Remote Sensing, IF 5.3 (2022–2023, 2023–2024); Computer Modeling in Eng. & Sciences, IF 2.4 (2022–2023); Computers, Materials & Continua, IF 3.1 (2022–2024, two editions); Frontiers in Communications & Networks (2023–2024); Discover Internet of Things, Springer (2024–2025).
General Chair: SmartTech 2024 (Morocco); IPCAINT 2024 (Morocco); SmartTech 2022 (Saudi Arabia).
Special Session Organizer/Chair: IJCNN 2025 & 2026, IRICT 2025, IWCMC 2023–2025, KES 2023–2026, ICCCI 2022–2024.
Reviewer: IEEE TGRS, IEEE TCE, AI Review, Neural Comp. & Apps., CVIU, KAIS, PAA, IEEE Access, Remote Sensing, Sensors.

Teaching & Educational Leadership

Outstanding Teacher Award, CCIS, Prince Sultan University (2025). Introduced computational-thinking-driven teaching and technology-enhanced learning practices.
Curriculum leadership: Core member, Academic Committee for M.Sc. & B.Sc. in AI at PSU (2023–2026). Designed and introduced Machine Learning and Big Data Analytics as new B.Sc. offerings at PSU and Taibah University, embedding industry-relevant tools and live research datasets. Pedagogical approach emphasizes computational thinking, integrating IT credentialing (badges/certificates) and technology-mediated play to enhance student engagement.
Courses taught: Machine Learning, Big Data Analytics, AI, Deep Learning, Digital Logic, Advanced Java, Web Development (HTML5/ASP.NET/MVC), Database Administration, Cryptography, Software Engineering, OOP, Data Fusion, Computer Networks, Algorithms & Data Structures.
Industry training: 200+ professional sessions (30h each) on BI, databases, and web development for banking, government, and NGO sectors in Tunisia and Saudi Arabia.

Industry & Translational Impact

Translational work spans national-scale BI deployment, patented AI systems for precision agriculture, and applied healthcare sensing—sustained over 15+ years across government, banking, and NGO sectors.
Prior industry affiliations: Senior Consultant, Sungard (US, 2011–2015); Senior Consultant, Business & Decision (France, 2013–2015); Senior Consultant, GIZ GmbH (Germany, 2009–2011). Over 200 professional training sessions delivered.

1. **National BI Deployment (Tunisia, 2009–2015):** As Senior Consultant for Sungard (US) and Business & Decision (France), designed BI solutions deployed across 174 training centers of three national agencies (ATFP, ONTT, AVFA) and social security funds, improving data-driven workforce planning at national scale.

2. **Red Palm Weevil Detection (Saudi Arabia, 2022–present):** Translated DL research into a patented acoustic+UAV system (SAIP SA 18938 B1) for early RPW detection. Under commercial field evaluation, addressing a pest costing the Gulf date industry hundreds of millions annually.
3. **Web Portal and Digital Platform Development Projects (Tunisia, 2009–2015):** Designed and developed several web portals and websites for governmental and sectoral organizations in Tunisia.

Books

[1] **Boulila, W.**, Ammar, A., Abdelkader, M., et al. (2026). *The Rise of Agentic Intelligence: From Theory to Real-World Applications*. Springer.

[2] Ahmad, J., Latif, S., **Boulila, W.**, et al. (2025). *Edge Intelligence: Advanced Deep Transfer Learning for IoT Security*. Elsevier.

[3] Koubaa, A., Ammar, A., Ghouti, L., **Boulila, W.**, et al. (2025). *Generative AI and Large Language Models*. SCI 1214, Springer.

[4] **Boulila, W.**, Ahmad, J., Koubaa, A., et al. (2024). *Decision Making & Security Risk Management for IoT Environments*. AIS 106, Springer.

[5] **Boulila, W.** (2013). *Prédiction des changements en imagerie satellitaire*. Presses Académiques Francophones.

Selected Journal Publications

160+ total; 110+ Q1. Full list via profiles.

IF $\geq$ 10 (Selected)

[1] Attitallah, S., **Boulila, W.**, et al. (2020). DL & IoT for smart cities. *Comp. Sci. Rev.* **IF 12.9**

[2] Hajjaji, Y., **Boulila, W.**, et al. (2021). Big Data & IoT for environment. *Comp. Sci. Rev.* **IF 12.9**

[3] Jemmali, M., et al. (2022). Data-emergency scheduling. *IEEE TII*. **IF 11.6**

[4] Khazri, M., **Boulila, W.**, Farah, I.R. (2023). Graph DL for remote sensing. *Comp. Sci. Rev.* **IF 12.9**

[5] Varone, G., **Boulila, W.**, et al. (2024). CNN fusion for IoMT-enabled BCI. *Information Fusion*. **IF 18.6**

[6] Lv, J., **Boulila, W.**, et al. (2025). Multimodal speech for healthcare. *IEEE JSTSP*. **IF 13.7**

[7] Benjdira, B., **Boulila, W.**, et al. (2025). LLMs + diffusion models survey. *Comp. Sci. Rev.* **IF 12.7**

[8] Sundaram, G., **Boulila, W.** (2025). Multimodal GNN for consumer electronics. *IEEE TCE*. **IF 10.9**

[9] Siddique, A., **Boulila, W.** (2025). Privacy-Enhanced Pneumonia Diagnosis. *IEEE TCE*. **IF 10.9**

[10] Jarraya, I., et al. (2025). GNSS-denied UAV navigation. *Satellite Navigation*. **IF 10.1**

IF 7–10 (Selected)

[1] Safaei, M., Sundararajan, E. A., Driss, M., **Boulila, W.**, et al. (2021). ML for obesity prediction. *Computers in biology and medicine*. **Cited 1000+**

[2] **Boulila, W.**, et al. (2021). RS-DCNN for big RS data. *CEA*. **IF 8.6**

[3] **Boulila, W.**, et al. (2021). CNN-LSTM for land cover change. *Ecol. Informatics*. **IF 7.3**

[4] Rehman, M. U., et al. (2022). Privacy-Preserving Deep Learning for Cancer Diagnosis. *IEEE TNSE* **IF 7.9**

[5] Jemmali, M., et al. (2022). Drone transportation for solar monitoring. *IEEE TITS*. **IF 9.5**

[6] **Boulila, W.**, et al. (2023). RPW detection via acoustic DL. *CEA*. **IF 8.6**

[7] Melhim, L., et al. (2024). Drone-assisted forest conservation. *IEEE IoT J.* **IF 8.2**

[8] **Boulila, W.**, et al. (2024). Weight initialization for satellite DL. *ESWA*. **IF 7.5**

[9] Shahid, L., **Boulila, W.**, et al. (2024). DTL-IDS intrusion detection. *JNCA*. **IF 8.7**

[10] Ali, A.M., et al. (2025). Dehazing-DiffGAN for RS. *IEEE TGRS*. **IF 8.6**

[11] Wang, W., et al. (2025). Privacy-preserving federated learning. *IEEE TITS*. **IF 8.4**

[12] Zhao, D., et al. (2025). Performance Analysis for Mobile Systems. *IEEE TVT*. **IF 7.1**

Additional Information

Languages: Arabic (Native), French (Fluent), English (Fluent), Italian (Working Proficiency)
Certifications: MCSA: SQL Server 2012; MCTS: SQL Server 2008; Microsoft Azure Fundamentals; IBM Big Data Engineer—Mastery Award (2018)